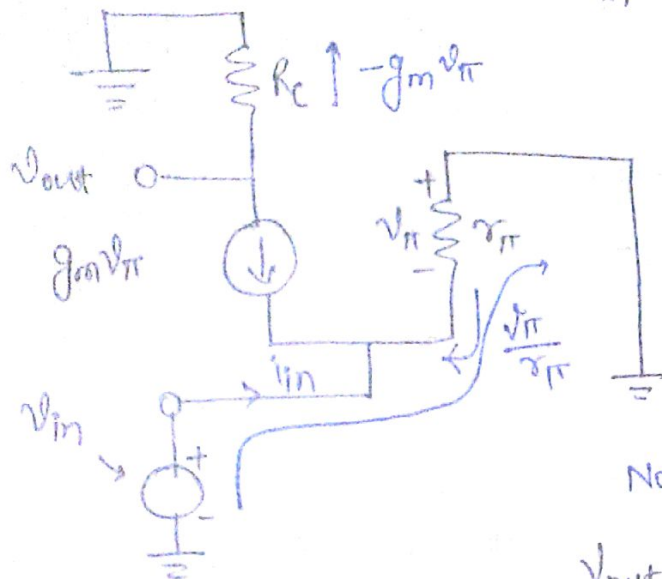
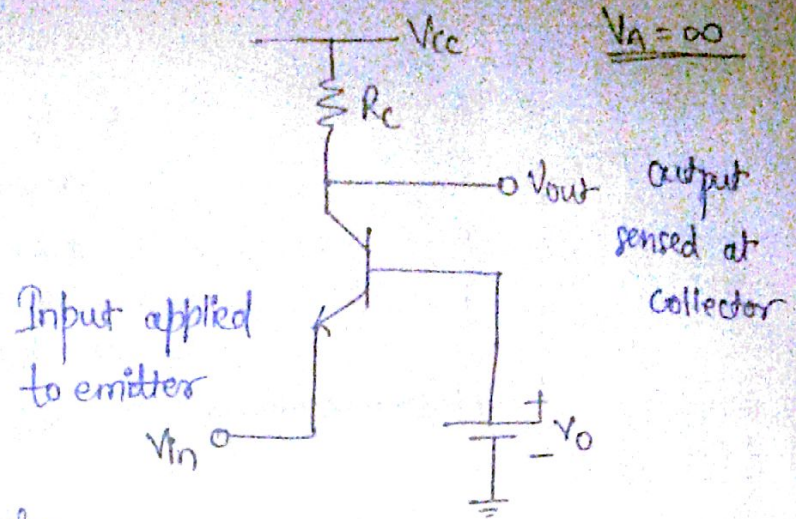


Lecture-26

CB stage:-

For this topology, we can not use previous results.



$$\therefore v_{\pi} = -v_{in}$$

Now, we write KCL at node output

$$V_{out} = -g_m v_{\pi} R_c = -g_m (-v_{in}) R_c$$

$$\Rightarrow \boxed{A_v = \frac{V_{out}}{V_{in}} = g_m R_c} \quad \text{Voltage Gain CB stage}$$

→ It is just same as gain of CE stage except the negative sign.

→ Now for input impedance,

$$i_{in} = g_m v_{\pi} + \frac{v_{\pi}}{r_{\pi}} = 0$$

$$i_{in} = v_{in} \left(g_m + \frac{1}{r_{\pi}} \right) = v_{in} \left(\frac{1 + g_m r_{\pi}}{r_{\pi}} \right) = v_{in} \left(\frac{1 + \beta}{r_{\pi}} \right)$$

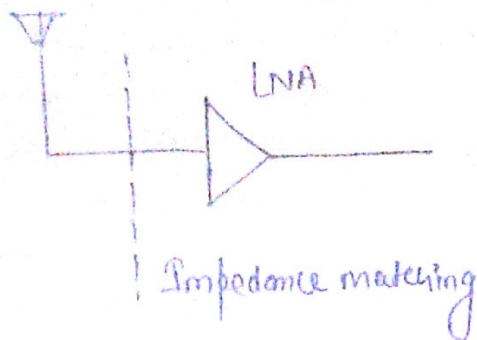
$$\approx v_{in} \left(\frac{\beta}{r_{\pi}} \right) \approx v_{in} g_m$$

$$\boxed{R_{in} \approx \frac{1}{g_m}}$$

Input impedance of CB stage

the input impedance of CB stage is β times lower than input impedance of CE stage.

* Antenna - LNA interface -



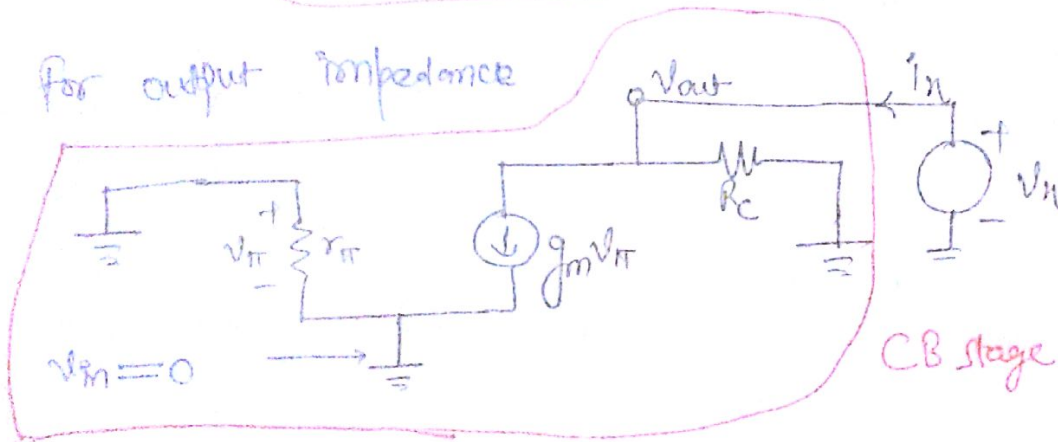
* If we connect an antenna to other circuit then we need there impedance matching.

* The impedance matching means impedance of LNA must be conjugate match of impedance of Antenna.

* According to Impedance matching

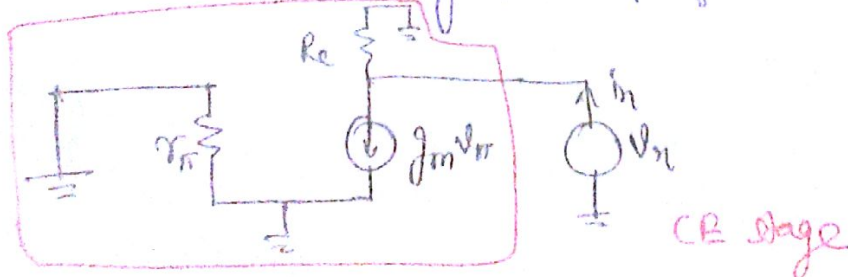
$$R_{in}|_{LNA} = R_{in}|_{Antenna}$$

* For output impedance



CB stage

Now we draw small signal model for CB stage



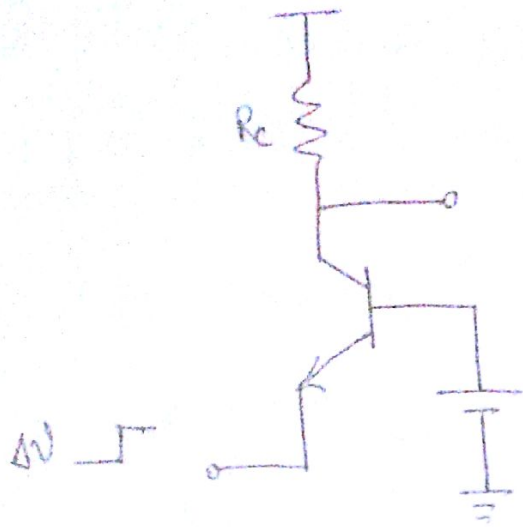
CB stage

So both circuits are same because when we are calculating the output impedance of the circuit we don't know what input is, so we set it to zero.

$$R_{out} = R_c$$

Question:- Why doesn't the CB stage invert?

It means why voltage gain is positive.



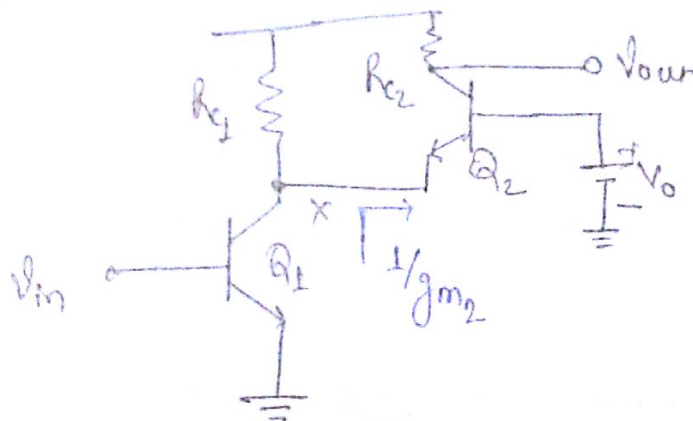
* When the emitter voltage goes from lower value to higher, since base voltage is fixed therefore V_{BE} decreases.

* As the result of decrease in V_{BE} I_C also decreases. As the I_C decreases V_{out} increases.

$$V_{out} = V_{CC} - I_C R_C$$

Therefore, as we increase the input signal output signal also increases that means it does not have inversion.

Quiz:-



find A_v ?

Soln:-

$$A_v = \frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_x} \cdot \frac{V_x}{V_{in}}$$

$$\therefore \frac{V_x}{V_{in}} = \text{Voltage gain of CE stage}$$

$$= -g_m \times (\text{total resistance b/w collector and ac ground})$$

$$= -g_m (R_{C1} \parallel \frac{1}{g_{m2}})$$

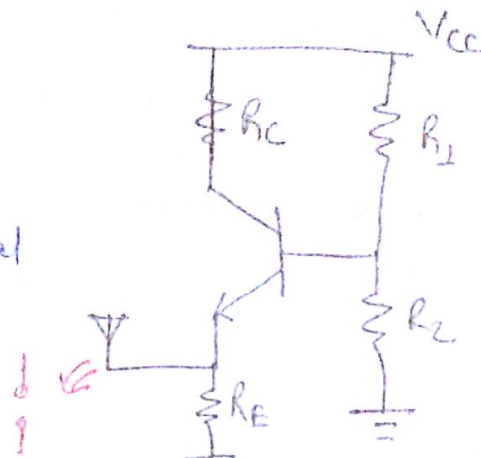
$$\frac{V_{out}}{V_x} = \text{Voltage gain of CB stage}$$

$$= g_{m2} \times R_{C2}$$

$$A_v = -g_{m1} \left(R_{C1} \parallel \frac{1}{g_{m2}} \right) (g_{m2} R_{C2})$$

* Bias Design :-

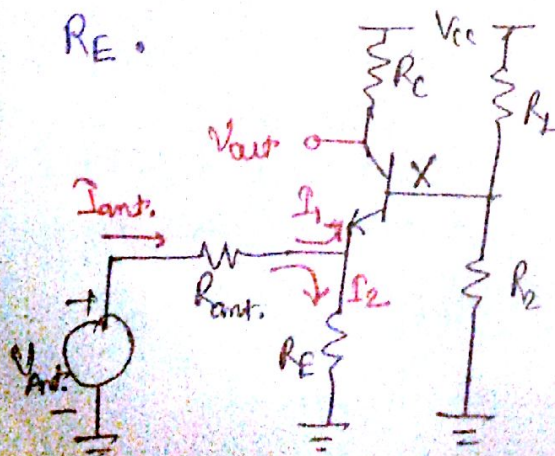
→ If we connect an antenna for receiving signal to emitter of CB stage, then it generates an electrical signal which goes to emitter of CB stage.



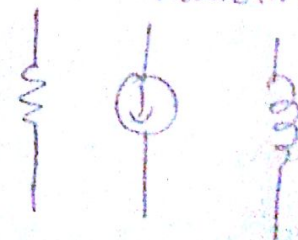
→ If there is no received signal then we do not have any voltage at emitter. Since the antenna is open circuit as far as dc current is concerned, so this ~~antenna~~ device is incapable of carrying dc current.

→ Antenna does not allow to flow dc current from emitter, so for dc current conduction we can use resistors, current sources and inductors. Therefore we take resistor

R_E .



DC current conduction:



→ Current is divided from I_{ant} into upward and downward directions. In the downward direction current gets wasted so avoiding to be wasted we take R_E as high resistance so that minimum current flows through it.

The current in upward direction goes through transistor and generates V_{out} .

→ The current I_1 and I_2 is given by

$$I_1 = \frac{R_E}{R_E + \frac{1}{g_m}} I_{ant.}$$

$$I_2 = \frac{\frac{1}{g_m}}{R_E + \frac{1}{g_m}} I_{ant.}$$

For avoiding I_2 to be wasted we maximize current I_1 , for that $R_E \gg \frac{1}{g_m}$ so that approximately all of the current goes upward.

Example:

$$V_{CC} = 1.5V, P = 1.5mW, \beta = 100, I_C \approx 1mA$$

$$\frac{1}{g_m} = 26\Omega \Rightarrow R_E = 260\Omega \Rightarrow V_E = 260mV \approx I_C R_E$$

($R_E = 10/g_m$)

$$V_X = V_{BE} + V_E \approx 1.06V$$

$$\frac{V_{CC}}{R_1 + R_2} = 100\mu A$$

$$R_1 = 10.6k\Omega, R_2 = 4.9k\Omega$$

$$\Rightarrow \frac{R_2}{R_1 + R_2} V_{CC} = 1.06V$$

Since $A_{voc} R_C$ so R_C should be max.

$$V_C \geq V_X$$

$$\Rightarrow V_{CC} - I_C R_C \geq V_X (=1.06V)$$

$$\Rightarrow R_C = 440\Omega$$

If $f_E = \frac{20}{g_m} = 520 \Omega$, $\Rightarrow V_E = 520 \text{ mV}$

$V_X = V_{BE} + V_E = 1.32 \text{ V}$

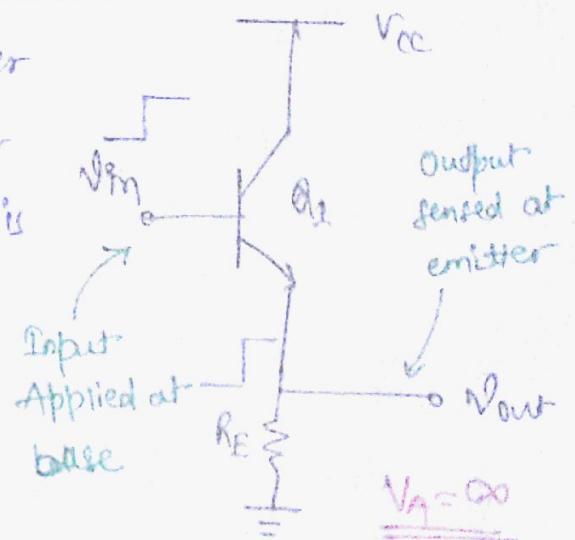
$V_{CC} - I_E R_E \geq 1.32$

$R_E \downarrow$ therefore $A_v \downarrow$ because $A_v = g_m R_E$. So for increasing R_E we lose gain.

Lecture - 27

• Emitter Follower (Common collector) :-

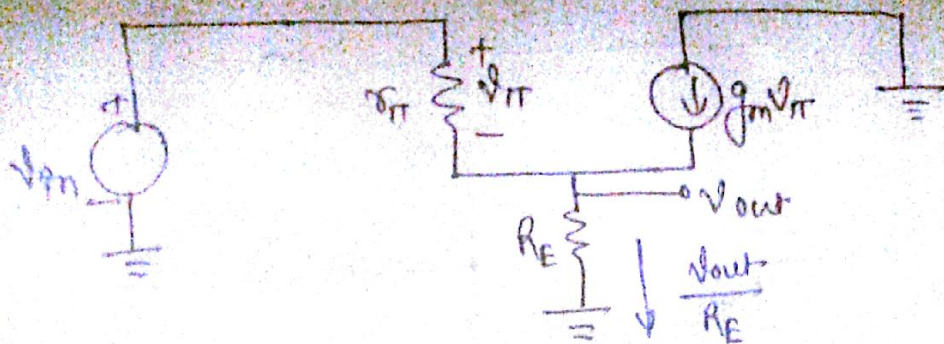
→ In this R_E is taken at emitter because we don't have a path for bias current to flow (+ve to -ve is not present, since +ve is present but -ve is not present here).



→ Assume device is biased properly

then we increase base input voltage then V_E also increase as the current I_E increases. That means, if input goes up then output also goes up. The output does not go up as much as the input so the gain is +ve and less than 1. This

device is called emitter follower because output follows the input, that is, if the input goes up output goes up or vice versa.



Av:-

KVL in input loop,

$$\Rightarrow v_{\pi} = v_{in} - v_{out}$$

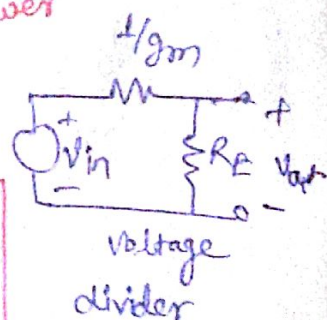
KCL at emitter node,

$$\frac{v_{in} - v_{out}}{r_{\pi}} + g_m(v_{in} - v_{out}) = \frac{v_{out}}{R_E}$$

$$v_{in} \left(\frac{1}{r_{\pi}} + g_m \right) = v_{out} \left(\frac{1}{R_E} + \frac{1}{r_{\pi}} + g_m \right)$$

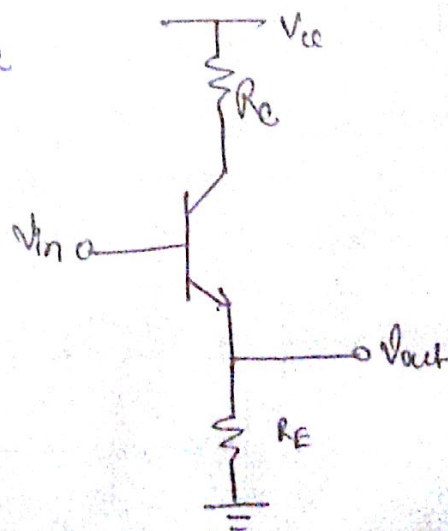
$$\Rightarrow \boxed{\frac{v_{out}}{v_{in}} = \frac{R_E}{\frac{1}{g_m} + R_E}} \quad \leftarrow \text{Voltage gain of Emitter follower}$$

$$A_v = \frac{\text{Resistance tied between emitter \& ground}}{\frac{1}{g_m} + \text{Resistance tied between emitter \& ground}}$$

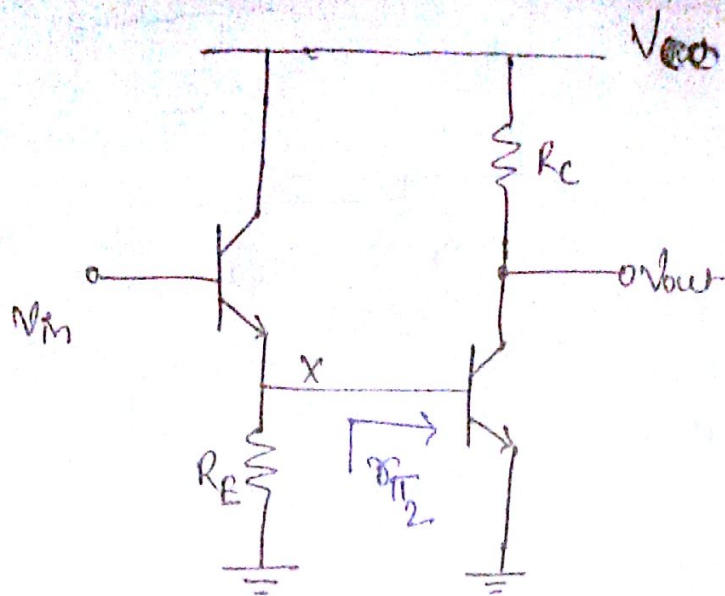


Question:- Do these results change if we insert a resistor in collector?

No, these results does not change because R_C does not have any effect on the gain.



Example:-



$\frac{V_x}{V_{in}}$ = voltage gain of emitter follower

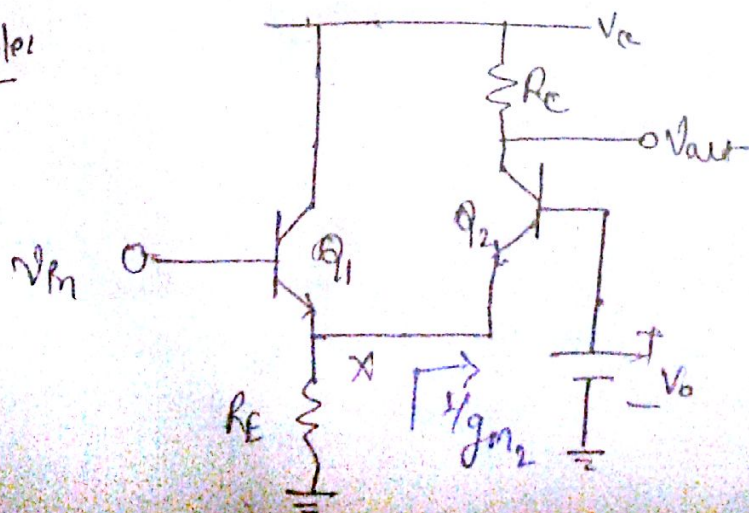
$$\frac{V_x}{V_{in}} = \frac{R_E \parallel r_{\pi 2}}{\frac{1}{g_{m1}} + (R_E \parallel r_{\pi 2})} < 1$$

$\frac{V_{out}}{V_x}$ = gain of CE stage

$$= -g_{m2} R_C$$

$$\Rightarrow A_v = \frac{V_{out}}{V_{in}} = -g_{m2} R_C \left(\frac{R_E \parallel r_{\pi 2}}{\frac{1}{g_{m1}} + (R_E \parallel r_{\pi 2})} \right)$$

Example:-



$\frac{V_{out}}{V_{in}}$ = Voltage gain of emitter follower

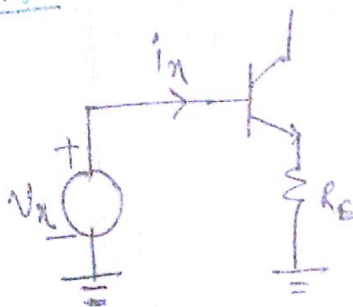
$$= \frac{R_E \parallel \frac{1}{g_{m2}}}{\frac{1}{g_{m1}} + \left(R_E \parallel \frac{1}{g_{m2}}\right)} < 1$$

$\frac{V_{out}}{V_{in}}$ = Voltage gain of CB stage

$$= g_{m2} R_C$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{\left(R_E \parallel \frac{1}{g_{m2}}\right)(g_{m2} R_C)}{\left(\frac{1}{g_{m1}}\right) + \left(R_E \parallel \frac{1}{g_{m2}}\right)}$$

R_{in} :

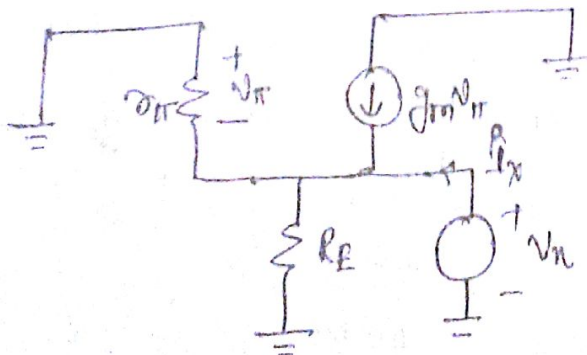


$$R_{in} = r_{\pi} + (\beta + 1) R_E$$

Same as CE stage

← with emitter degeneration

R_{out} :



→ Since R_E is in parallel with V_n so we first find rest of the circuit after removing R_E from it.

$$\therefore V_{\pi} = -V_n$$

$$\therefore \frac{-V_n}{r_{\pi}} - g_m V_{\pi} + I_n = 0$$

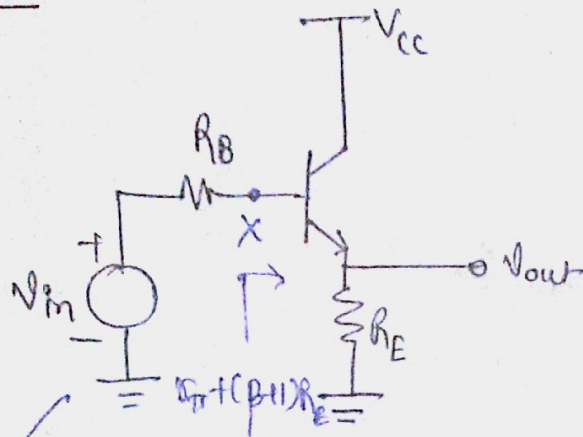
$$\therefore \frac{V_n}{I_n} \approx \frac{1}{g_m}$$

$$\Rightarrow \boxed{R_{out} = \frac{1}{g_m} \parallel R_E}$$

output impedance of
Emitter follower

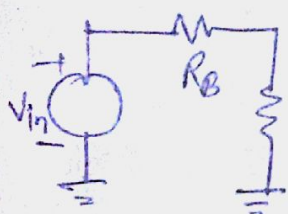
It is same as CB stage because R_C does not affect emitter follower.

Example:-



$$\frac{V_{in}}{V_{in}} = \frac{[r_{\pi} + (\beta+1)R_E]}{R_B + [r_{\pi} + (\beta+1)R_E]}$$

$$\frac{V_{out}}{V_{in}} = \frac{R_E}{\frac{1}{g_m} + R_E}$$

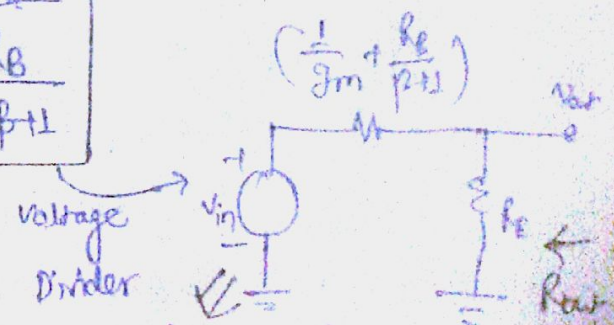


$$A_v = \left(\frac{R_E}{R_E + \frac{1}{g_m}} \right) \left[\frac{r_{\pi} + (\beta+1)R_E}{R_B + [r_{\pi} + (\beta+1)R_E]} \right]$$

$$\approx \frac{\left(\frac{1}{g_m} + R_E \right)}{\left[\frac{1}{g_m} + R_E + \frac{R_B}{(\beta+1)} \right]} \cdot \frac{R_E}{\left(R_E + \frac{1}{g_m} \right)}$$

$$\boxed{A_v = \frac{R_E}{\frac{1}{g_m} + R_E + \frac{R_B}{\beta+1}}}$$

$$\boxed{R_{in} = R_B + r_{\pi} + (\beta+1)R_E}$$



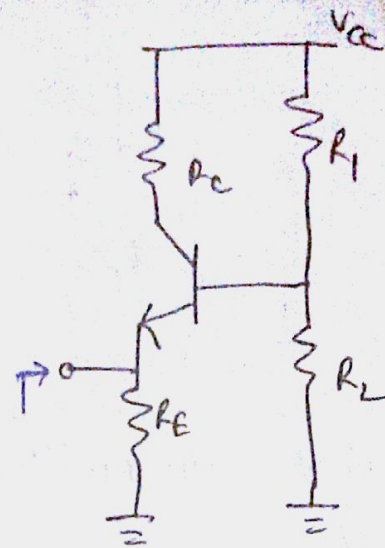
$$\boxed{R_{out} = R_E \parallel \left(\frac{R_B}{\beta+1} + \frac{1}{g_m} \right)}$$

Lecture - 28

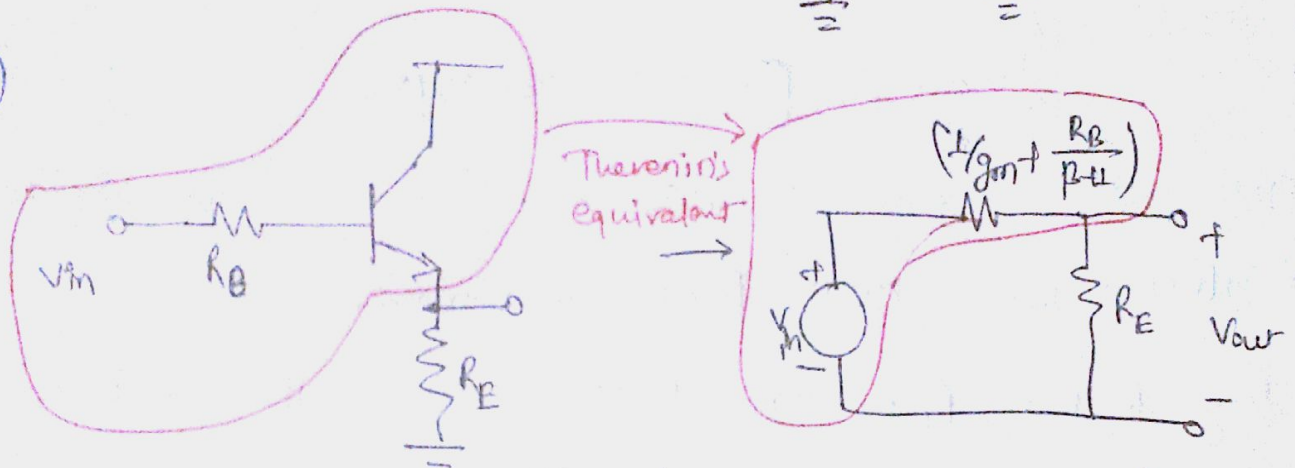
Two Remarks:-

①

$$R_{in} = R_E \parallel \left(\frac{1}{g_m} + \frac{R_1 + R_2}{\beta + 1} \right)$$

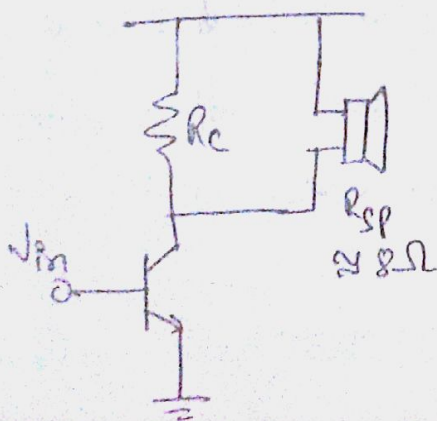


②



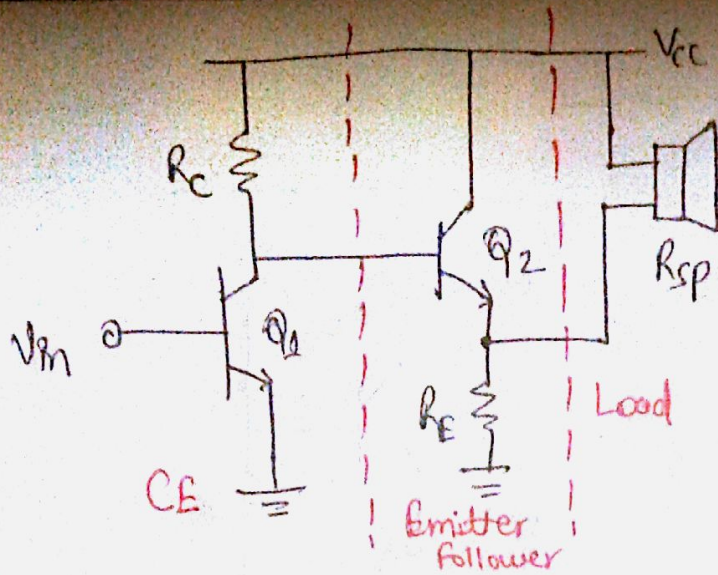
We can not use this equivalent ckt for finding R_{in} because this is the equivalent Thevenin's circuit of transistor.

* Emitter follower as a buffer:-



$$A_v = -g_m (R_c \parallel R_{sp})$$

So with the R_{sp} gain reduces by an large amount because of R_{sp} . For this we use emitter follower.



$$V_A = \infty$$

→ We interpose an emitter follower between CE and loudspeaker.

$$\frac{V_{out}}{V_{in}} = -g_m \left[R_c \parallel r_{\pi} + \underbrace{(\beta+1)(R_E \parallel R_{sp})}_{\approx 800\Omega} \right] \approx 8\Omega$$

This tells us that by this configuration voltage gain does not affected much as compared to previous.

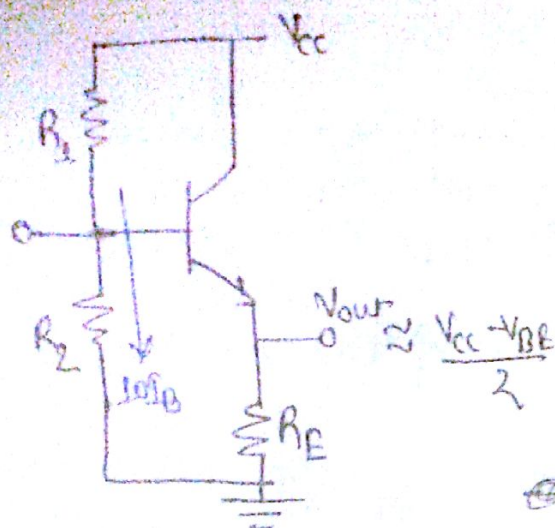
Say $\beta=100$, $R_E = 500\Omega$ therefore $r_{\pi} + (\beta+1)(R_E \parallel R_{sp})$ is quite larger than R_{sp} so it is preferable.

$$\frac{V_{out}}{V_{in}} \approx \frac{R_E \parallel R_{sp}}{R_E \parallel R_{sp} + \frac{1}{g_{m2}}} < 1$$

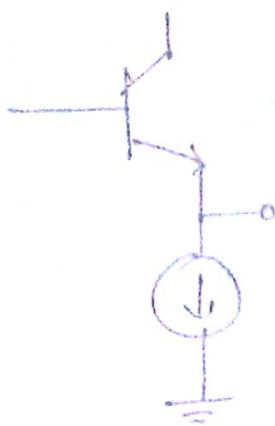
Therefore, when we calculate A_v then it is approximately equal to the gain of CE stage and quite higher than simple BJT stage with speaker. Since Emitter follower does not affect voltage gain so it acts like a buffer. Therefore we ~~lose~~ lose less signal than previous.



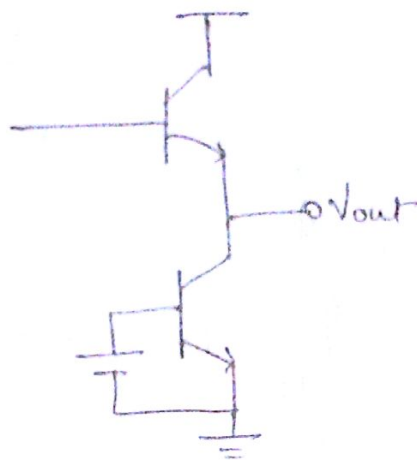
* Base Design -



→ This device does not saturate in any condition because at collector voltage is V_{CC} and at base it is in divided form, so in always condition $V_{CE} > V_{BE}$.



⇒



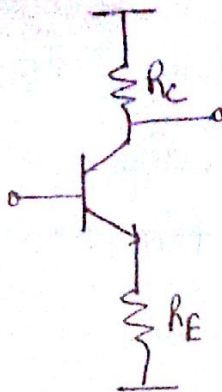
→ At the emitter, it will work as constant current source approximately which can be designed by transistor as constant ~~current~~ current source.

* If Early voltage V_A is finite then emitter follower is designed in small signal equivalent we get r_o across collector and emitter. Since ~~collector~~ collector is at ac ground so top terminal of r_o is at ground and bottom terminal will be connected to emitter therefore r_o will be parallel with R_E . So in previous result of emitter follower, we just replace R_E by $(r_o || R_E)$.

When $V_A < \infty$ in emitter follower

CE stage:-

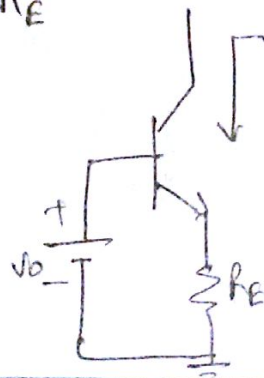
$$V_A = \infty$$



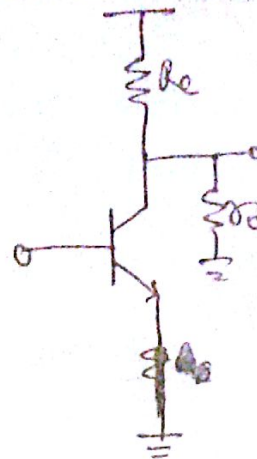
$$A_v = - \frac{R_c}{\frac{1}{g_m} + R_E}$$

$$R_{in} = r_{\pi} + (\beta + 1) R_E$$

$$R_{out} = R_c$$



$$V < \infty$$



$$A_v = -g_m (R_c \parallel r_o)$$

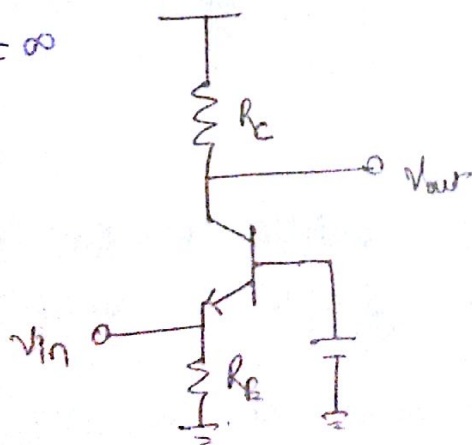
$$R_{in} = r_{\pi}$$

$$R_{out} = R_c \parallel r_o$$

$$R_{out} = (1 + g_m r_o) (R_E \parallel r_o) + r_o$$

CB stage:-

$$V_A = \infty$$



$$A_v = g_m R_c$$

$$R_{in} = \frac{1}{g_m} \parallel R_E$$

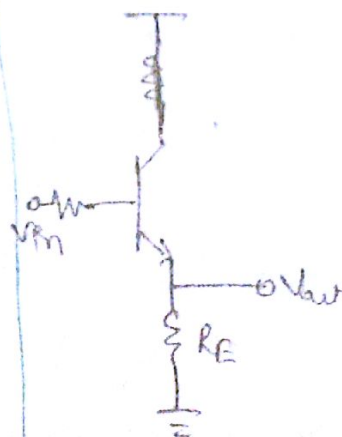
$$R_{out} = R_c$$

Emitter follower:-

$$V_A = \infty$$

(for $V_A < \infty$

$$R_E \rightarrow R_E \parallel r_o)$$

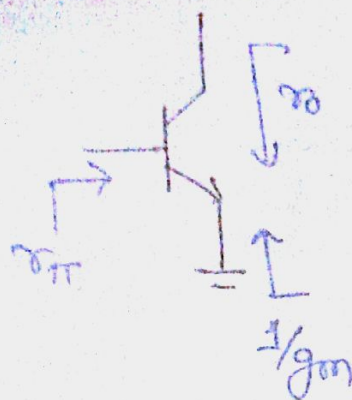


$$A_v = \frac{R_E}{R_E + \frac{1}{g_m} + \frac{R_B}{\beta + 1}}$$

$$R_{in} = R_B + r_{\pi} + (\beta + 1) R_E$$

$$R_{out} = R_E \parallel \left(\frac{R_B}{\beta + 1} + \frac{1}{g_m} \right)$$

Useful Impedance Expressions:-

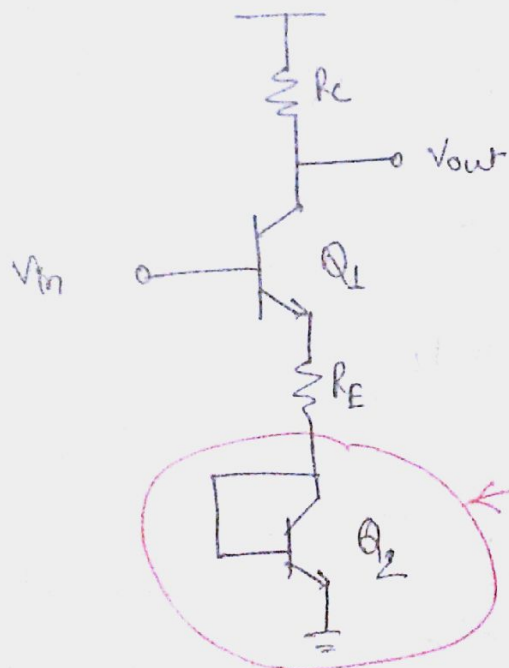


→ If the emitter of the bipolar transistor is at ac ground or any ground then looking into the base we see r_{π} .

→ If emitter and base of the bipolar transistor is at ground then into the collector, we see r_o .

→ If $V_A = \infty$ and base is at ground then looking into the emitter, we see $1/g_m$ resistance.

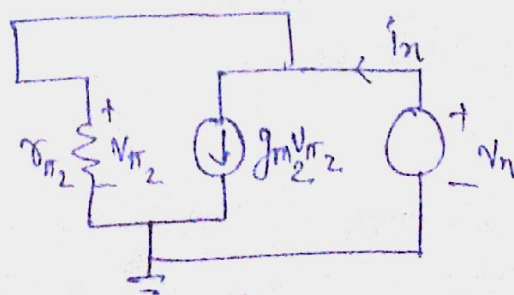
Quiz:



find Voltage gain?

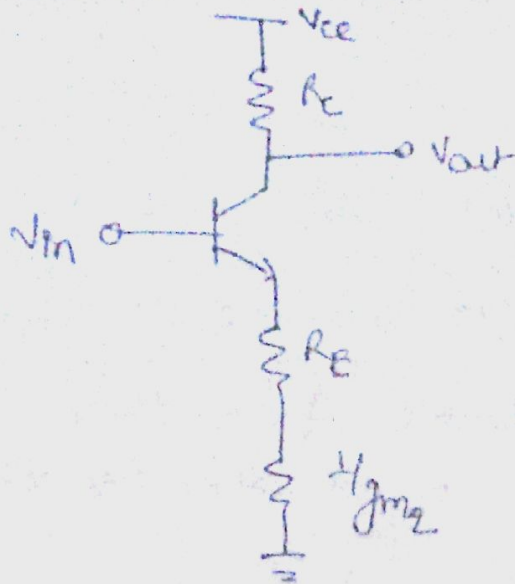
Diode connected Transistor

Soln:



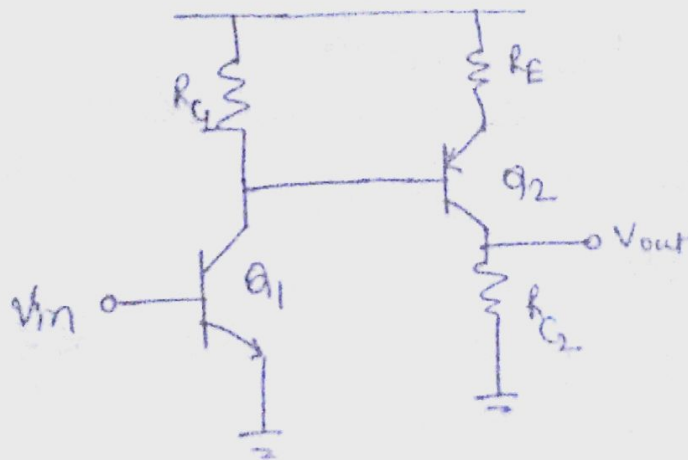
$$i_{n2} = \frac{v_n}{r_{\pi 2}} + g_m v_n$$

$$\Rightarrow \frac{v_n}{i_n} = \frac{1}{g_m} = \frac{V_T}{I_E}$$



$$A_v = \frac{R_C}{\frac{1}{g_{m1}} + R_E + \frac{1}{g_{m2}}}$$

Example 1



$$\frac{v_n}{v_{in}} = -g_{m1} \left[R_{C1} \parallel (R_{E2} + (\beta+1)R_{E1}) \right]$$

$$\frac{v_{out}}{v_n} = - \frac{R_{C2}}{R_{E2} + \frac{1}{g_{m2}}}$$

$$A_v = \frac{v_{out}}{v_{in}} = \left(\frac{R_{C2}}{R_{E2} + \frac{1}{g_{m2}}} \right) \left(-g_{m1} \left[R_{C1} \parallel (R_{E2} + (\beta+1)R_{E1}) \right] \right)$$